

Crossing-Point Analysis for Weighted Rolling Validation: Theory

Aileen Tao

1 Question and approach

Rolling validation (RV) selects between models by adding their validation losses over successive training sizes. The target, however, is the model with smaller population prediction risk at the *current* training size. If that preferred model changes as the training sample grows, the two comparisons need not change sign together: RV still includes losses from earlier, smaller training samples. The question of this report is whether weighting later validation observations can bring the expected RV crossing close to the population-risk crossing.

The two tasks are distinct. First, expected RV should change preference at approximately the right training size (*alignment*). Second, the realized, random RV comparison should be sufficiently concentrated around its mean (*concentration*). Weighting can improve the first task while making the second harder.

Why Analysis 08 is not enough

Analysis 08 considered a sample mean against an oracle that was always better in population risk. It studied the mean and variance of the RV loss difference and whether their ratio gave a selection-consistency bound. There was no preference reversal, hence no crossing to align. The present model deliberately has a reversal. Its nonzero mean parameter also creates a new linear random term in RV, so its variance cannot be imported from 08 without calculation.

Question	Quantity	Main result	Proof
Population crossing	$\Delta_R(n)$	$n_{\text{pop}} = \lfloor \theta^{-2} \rfloor + 1$	A.1
Ordinary RV lag	$\Delta_{\text{RV}}(n)$	$n_{\text{RV}} \sim \theta^{-2} \log(\theta^{-2})$	A.3
Weight alignment	A_n/W_n	$nA_n/W_n \rightarrow 1$ is sufficient	B.4
Logarithmic weights	A_n/W_n	No	B.5, B.6
Geometric weights	A_n/W_n	Yes	B.7
Near-crossing tradeoff	variance/mean ²	No under the 09 conditions	C.3, D.2

2 The two comparisons studied in 09

For models 0 and 1 trained on n observations, write

$$\Delta_R(n) = R_0(n) - R_1(n).$$

A negative value favors Model 0, a positive value favors Model 1, and zero means equal population risks. If $RV_{k,w}(n)$ denotes Model k 's weighted RV loss through the validation time with n training observations, write

$$\Delta_w(n) = \mathbb{E}[RV_{0,w}(n) - RV_{1,w}(n)].$$

Its sign describes the preference of *expected* weighted RV, not necessarily a particular realized selection. Thus $\Delta_R(n)$ tells us what the population comparison prefers, while $\Delta_w(n)$ tells us what expected weighted RV prefers. We compare their first strict negative indices:

$$n_{\text{pop}} = \min\{n \geq 1 : \Delta_R(n) < 0\}, \quad n_{\text{RV},w} = \min\{n \geq 1 : \Delta_w(n) < 0\}.$$

The sets are shown to be nonempty below for the weights used here. The goal is to make these indices close, principally in the ratio sense as the crossing moves outward.

3 A model with a genuine crossing

Let

$$Y_i = \theta + \xi_i, \quad \xi_i \stackrel{\text{iid}}{\sim} N(0, 1), \quad 0 < |\theta| < 1.$$

Model 0 predicts a new response using the training sample mean $\bar{Y}_n$; Model 1 always predicts zero. Both use squared loss. Model 0 is unbiased but has estimation variance. Model 1 has no estimation variance but retains squared bias θ^2 . Hence the zero predictor can be better early, and the sample mean better later. A fixed shrinkage predictor would give a similar bias–variance crossing, but the zero predictor keeps the mechanism and formulas transparent.

3.1 Population prediction risk

Here $R_k(n)$ averages squared prediction loss over the training data and an independent new response. The exact crossing is as follows.

Theorem 3.1 (Population crossing).

$$R_0(n) = 1 + \frac{1}{n}, \quad R_1(n) = 1 + \theta^2, \quad \Delta_R(n) = \frac{1}{n} - \theta^2, \quad n_{\text{pop}} = \lfloor \theta^{-2} \rfloor + 1.$$

The common 1 is new-response noise. The remaining comparison is the sample mean’s $1/n$ estimation variance against the zero predictor’s permanent θ^2 bias. The risks tie at $n = \theta^{-2}$ when that is an integer; the displayed index is the first *strict* Model-0 preference. The calculation and equality case are proved in Appendix A.1.

3.2 Why ordinary RV crosses late

At validation time $m + 1$, Model 0 has only m training observations. Its expected loss difference from Model 1 is $1/m - \theta^2$: positive early, negative after $m > \theta^{-2}$. Ordinary RV sums *all* these contributions,

$$\Delta_{\text{RV}}(n) = \sum_{m=1}^n \left(\frac{1}{m} - \theta^2 \right) = H_n - n\theta^2, \quad H_n = \sum_{m=1}^n \frac{1}{m}. \quad (1)$$

Even after current population risk favors Model 0, RV remembers the positive losses from early, noisy sample means. This delays its crossing.

Proposition 3.2 (Ordinary-RV lag). *As $\theta \rightarrow 0$,*

$$n_{\text{RV}} \sim \theta^{-2} \log(\theta^{-2}), \quad \frac{n_{\text{RV}}}{n_{\text{pop}}} \sim \log(\theta^{-2}) \rightarrow \infty.$$

The one-time expectation is derived in Appendix A.2; the crossing asymptotic and integer bracketing are proved in Appendix A.3.

4 Weighted RV and crossing alignment

The lag originates in accumulated early contributions. Increasing weights give more influence to later validations, which use larger training samples. For deterministic $w_i > 0$, independent of the data, define

$$\Xi_{n,w} = \sum_{i=2}^{n+1} w_i \{(Y_i - \bar{Y}_{i-1})^2 - Y_i^2\}, \quad \Delta_w(n) = \mathbb{E}[\Xi_{n,w}].$$

The final validation prediction uses n training observations. A negative statistic selects Model 0. Multiplying all weights by one positive constant changes neither sign nor crossing. The expected contribution at time i is $d_i = (i-1)^{-1} - \theta^2$. Define

$$A_n = \sum_{i=2}^{n+1} \frac{w_i}{i-1}, \quad W_n = \sum_{i=2}^{n+1} w_i.$$

Then

$$\Delta_w(n) = \sum_{i=2}^{n+1} w_i d_i = A_n - \theta^2 W_n. \quad (2)$$

This follows from the one-time loss calculation, without assuming validation losses are independent; Appendix B.1 gives the derivation.

Normalizing by W_n makes A_n/W_n the weighted average of the historical estimation-variance terms $1/(i-1)$. Population risk uses only the current value $1/n$. Their exact relationship is

$$\frac{\Delta_w(n)}{W_n} = \Delta_R(n) + \left(\frac{A_n}{W_n} - \frac{1}{n} \right). \quad (3)$$

The parenthesized correction is nonnegative, and strictly positive for $n > 1$ with positive weights. It is precisely the historical-training-size effect that keeps the expected RV comparison behind current population risk. In particular, expected RV cannot cross before the population criterion. The identity and strictness are proved in Appendix B.2; monotonicity and existence of the weighted crossing are proved in Appendix B.3.

To imitate population risk near its moving crossing, A_n/W_n should behave like $1/n$. This leads to a sufficient condition.

Theorem 4.1 (Sufficient crossing alignment). *If the positive deterministic weights do not depend on θ and*

$$\frac{nA_n}{W_n} \longrightarrow 1, \quad (4)$$

then $n_{\text{RV},w}$ exists for all sufficiently small nonzero θ and

$$\frac{n_{\text{RV},w}}{n_{\text{pop}}} \longrightarrow 1 \quad (\theta \rightarrow 0).$$

Interpretation: the weighted average of reciprocal training sizes is asymptotically the reciprocal of the latest training size. This aligns the *expected* sign changes; it says nothing yet about random fluctuations. The ε -bracketing proof is in Appendix B.4.

4.1 Representative weights

The following families illustrate the condition. For $w_i = [\log(i+1)]^p$, $p > 0$, Appendix B.5 proves

$$\frac{A_n}{W_n} \sim \frac{\log n}{(p+1)n}, \quad \frac{n_{\text{RV},w}}{n_{\text{pop}}} \sim \frac{\log(\theta^{-2})}{p+1}.$$

The cases $p = 1$ and $p = 2$ are the log and squared-log weights. Ordinary weights $w_i = 1$ are $p = 0$. For $w_i = \log \log(i+e)$, Appendix B.6 proves $A_n/W_n \sim (\log n)/n$ and the ordinary-RV leading crossing ratio. All four fail (4): they reduce finite lags to different degrees but remain asymptotically late.

Geometric weights $w_i = q^i$, $q > 1$, are a contrasting diagnostic. Appendix B.7 proves $nA_n/W_n \rightarrow 1$, so Theorem 4.1 aligns their crossing ratio. The same appendix calculates $\sum w_i^2/W_n^2 \rightarrow (q-1)/(q+1)$, showing how little averaging these weights retain.

5 Randomness near the moving crossing

Alignment of *expected* RV is only the first requirement. A realized RV statistic may still have the wrong sign. A standard sufficient concentration condition is

$$\frac{\text{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \rightarrow 0$$

when the mean is nonzero: Chebyshev's inequality then controls the probability of a sign error relative to that mean. We ask whether this condition can accompany (4).

The relevant limit is $\theta \rightarrow 0$ with $n\theta^2 \rightarrow c \in (0, \infty)$. Since $n_{\text{pop}} \sim \theta^{-2}$, c locates n relative to the moving population crossing: $c < 1$ is before, $c > 1$ after, and $c = 1$ at its leading-order location. Appendix C.1 gives the precise sign calculation.

The new 09 model has a linear Gaussian noise term absent when $\theta = 0$. The exact finite- n decomposition is

$$\text{Var}(\Xi_{n,w}) = 2 \sum_{t=1}^n a_t^2 + 4 \sum_{l=2}^{n+1} (l-1)b_l^2 + 4\theta^2 \sum_{i=2}^{n+1} w_i^2, \quad (5)$$

where the coefficient definitions and Gaussian orthogonality proof are in Appendix D.2. This is a new 09 identity, not an unverified transfer of the 08 variance formula.

The mechanism of the tradeoff is direct. Alignment puts nearly all normalized weight near the latest validation times. That makes A_n/W_n resemble $1/n$, but shrinks the effective number of validation observations. The linear noise term in (5) then prevents Chebyshev concentration.

Theorem 5.1 (Near-crossing obstruction). *Under (4),*

$$n \frac{\sum_{i=2}^{n+1} w_i^2}{W_n^2} \rightarrow \infty.$$

For every sequence $\theta \rightarrow 0$ and $n\theta^2 \rightarrow c \in (0, \infty) \setminus \{1\}$,

$$\frac{\text{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \rightarrow \infty. \quad (6)$$

Thus in this 09 model the same positive deterministic weighting behavior that aligns crossings precludes a vanishing Chebyshev ratio on either fixed side of the moving crossing. This does *not* establish a limiting wrong-selection probability. The full effective-weight and variance-lower-bound proof is in Appendix C.3; the variance identity it uses is proved in Appendix D.2. At $c = 1$, a zero mean makes the ratio undefined; the nonzero smaller-order case is treated in Appendix C.4.

The logarithmic and log-log families illustrate a different failure: Appendix D.3 gives their near-crossing ratio order $(\log n)^{-2}$, but Appendix B shows their expected comparison remains positive for every fixed c . For $c > 1$, this is concentration around the sign *opposite* to current population preference. Geometric weights align but have a diverging near-crossing ratio, by Appendix B.7 and Theorem 5.1. Neither family is uniformly better.

At fixed $\theta \neq 0$, $n \rightarrow \infty$ is a different regime. Appendix D.4 proves that $w_i = \exp(i^\beta)$, $0 < \beta < 1$, satisfies alignment and a vanishing fixed- θ variance ratio because its effective window grows like $n^{1-\beta}$. Theorem 5.1 still rules out a vanishing Chebyshev ratio near the moving crossing.

6 What 09 establishes

The population crossing occurs at approximately θ^{-2} ; ordinary RV crosses later by a logarithmic factor. Weighted RV differs from current population risk by the explicit historical term in (3). The condition (4) aligns expected crossings, but, in the same near-crossing regime, forces the Chebyshev ratio to diverge. Logarithmic weights retain averaging but not alignment; geometric weights align while concentrating weight too strongly. These statements concern expected comparisons and variance bounds, not a proved limiting selection-error probability.

7 What remains open

The obstruction is specific to this model, positive deterministic weights, and the near-crossing Chebyshev criterion. It does not identify the limiting wrong-selection probability under aligned weights. A direct probability analysis, another validation design, or a different crossing model could lead to a distinct answer. Fixed- θ concentration under intermediate growth does not settle the moving-crossing question.

A Population risk and ordinary RV

A.1 Population-risk formulas and strict crossing

Let $Y_{\text{new}} = \theta + \xi_{\text{new}}$ be independent of the training sample. Since $\bar{Y}_n = \theta + S_n/n$ with $S_n = \sum_{t=1}^n \xi_t$, the Model-0 error is $\xi_{\text{new}} - S_n/n$. Independence, zero means, and $\text{Var}(S_n) = n$ give $R_0(n) = 1 + 1/n$. Model 1 has error $\theta + \xi_{\text{new}}$, hence $R_1(n) = 1 + \theta^2$. Their difference is $1/n - \theta^2$, negative exactly when $n > \theta^{-2}$. The least integer strictly above θ^{-2} is $\lfloor \theta^{-2} \rfloor + 1$, including when θ^{-2} itself is an integer (the equality index is a tie). This proves Theorem 3.1.

A.2 One-time and ordinary-RV expectation

At time $m + 1$, $\bar{Y}_m = \theta + S_m/m$ and the fresh error of Model 0 is $\xi_{m+1} - S_m/m$; that of Model 1 is $\theta + \xi_{m+1}$. Expanding and using independence of ξ_{m+1} and S_m gives

$$\mathbb{E}[(Y_{m+1} - \bar{Y}_m)^2 - Y_{m+1}^2] = (1 + 1/m) - (1 + \theta^2) = 1/m - \theta^2.$$

Summing over $m = 1, \dots, n$ proves (1).

A.3 Ordinary-RV crossing asymptotics

Put $h = \theta^2$ and $N = h^{-1}$. The mean $H_n - nh$ has increments $1/(n+1) - h$, so it is initially increasing and eventually decreasing to $-\infty$; its first strict negative index exists and is unique. Integral bounds give $\log(n+1) \leq H_n \leq 1 + \log n$. Fix $\varepsilon \in (0, 1)$ and set

$$n_- = \lfloor (1 - \varepsilon)N \log N \rfloor, \quad n_+ = \lceil (1 + \varepsilon)N \log N \rceil.$$

Both tend to infinity; $\log n_{\pm} = \log N + \log \log N + O(1)$, while $n_{\pm}h = (1 \pm \varepsilon) \log N + o(1)$. Thus $H_{n_-} - n_-h = \varepsilon \log N + o(\log N) > 0$ and $H_{n_+} - n_+h = -\varepsilon \log N + o(\log N) < 0$. The crossing lies between n_- and n_+ . Divide by $N \log N$ and let $\varepsilon \downarrow 0$ to obtain $n_{\text{RV}} \sim N \log N$. Since $n_{\text{pop}} = N + O(1)$, Proposition 3.2 follows.

B Weighted RV and crossing alignment

B.1 Weighted expectation

Apply the one-time expectation in Appendix A.2 with $m = i - 1$. Deterministic weights and linearity of expectation give

$$\Delta_w(n) = \sum_{i=2}^{n+1} w_i ((i-1)^{-1} - \theta^2) = \sum_{i=2}^{n+1} \frac{w_i}{i-1} - \theta^2 \sum_{i=2}^{n+1} w_i = A_n - \theta^2 W_n.$$

No independence between validation losses is asserted or needed.

B.2 Exact remainder and crossing order

Divide the preceding identity by $W_n > 0$ and add and subtract $1/n$ to obtain (3). Each $1/(i-1) \geq 1/n$. Their positive-weighted average therefore satisfies $A_n/W_n \geq 1/n$; for $n > 1$, at least one earlier term is strictly larger and has positive weight, so the inequality is strict. If $\Delta_R(n) \geq 0$, then $\Delta_w(n)/W_n \geq 0$. Consequently a strict weighted-RV crossing cannot precede n_{pop} . Equality is possible at $n = 1$ or if nonnegative (rather than positive) weights are supported only on the final term.

B.3 Monotonicity and crossing existence

Write $r_n = A_n/W_n$. Adding the next validation time gives

$$r_{n+1} = \frac{W_n r_n + w_{n+2}/(n+1)}{W_n + w_{n+2}}.$$

Every term in r_n is at least $1/n > 1/(n+1)$, so $r_{n+1} < r_n$ for positive weights. Thus the equation $r_n < h$ has at most one first strict crossing. It exists exactly when $\inf_n r_n < h$; since r_n decreases, this is equivalent to $\lim_n r_n < h$. If the limit equals h , strict crossing need not occur. Under (4), $r_n \sim 1/n \rightarrow 0$, so crossing exists for every fixed $h > 0$; $r_1 = 1 > h$ because $0 < h < 1$.

B.4 Sufficient-alignment theorem

Let $h = \theta^2 \downarrow 0$ and fix $\varepsilon \in (0, 1)$. Set $n_- = \lfloor (1 - \varepsilon)/h \rfloor$ and $n_+ = \lceil (1 + \varepsilon)/h \rceil$. Both grow, and (4) gives $r_{n_{\pm}} = (1 + o(1))/n_{\pm}$. Therefore

$$\frac{r_{n_-}}{h} \rightarrow \frac{1}{1 - \varepsilon} > 1, \quad \frac{r_{n_+}}{h} \rightarrow \frac{1}{1 + \varepsilon} < 1.$$

By Appendix B.3, the first strict crossing lies in $(n_-, n_+]$ for small h . Rounding changes either endpoint by at most one. Divide by h^{-1} and let $\varepsilon \downarrow 0$; since $n_{\text{pop}} = h^{-1} + O(1)$, the crossing ratio tends to one. This proves Theorem 4.1.

B.5 Log-power weights and their crossings

Let $w_i = [\log(i+1)]^p$ with fixed $p \geq 0$. A shift by one changes $\log(i+1)$ by $O(1/i)$, so its p th power has the same leading order as $(\log i)^p$ (with $p = 0$ interpreted as 1). Integral comparison, or summation by parts, yields

$$A_n \sim \int_2^n \frac{(\log x)^p}{x} dx = \frac{(\log n)^{p+1}}{p+1}.$$

For completeness, the summand is eventually decreasing: its logarithmic derivative is $(p/\log x - 1)/x < 0$ for $x > e^p$. Thus its sum differs from the integral by a bounded initial contribution and an error no larger than a fixed eventual first term. The shift error is summable for $p \geq 0$. Also, for any $\delta \in (0, 1)$, the last $(1-\delta)n$ indices have weights $(\log n)^p(1+o(1))$ uniformly, while the first δn contribute at most $\delta n(\log n + O(1))^p$. Taking $n \rightarrow \infty$ and then $\delta \downarrow 0$ gives $W_n \sim n(\log n)^p$. Hence

$$r_n \sim \frac{\log n}{(p+1)n}.$$

For the crossing let $N = h^{-1}$ and $g_N = N \log N / (p+1)$. At $n_{\pm} = \lfloor (1 \mp \varepsilon)g_N \rfloor$ and $\lceil (1 \pm \varepsilon)g_N \rceil$, respectively, $\log n_{\pm} / \log N \rightarrow 1$, so $r_{n_{\pm}}/h \rightarrow (1 \mp \varepsilon)^{-1}$. Monotonicity from Appendix B.3 brackets the first crossing between them. Letting $\varepsilon \downarrow 0$ gives $n_{\text{RV},w} \sim g_N$ and, since $n_{\text{pop}} \sim N$, the ratio stated in the main text. This includes $p = 0$.

B.6 Log-log weight

Set $f(x) = \log \log(x+e) > 0$. Its derivative is $f'(x) = 1/[(x+e)\log(x+e)]$. For large x , $f(x) \sim \log \log x$ and $f(x)/x$ is decreasing, so integral comparison applies. The shift from $f(x)$ to $\log \log x$ produces an integrable error after division by x : the mean-value theorem bounds it by $C/(x^2 \log x)$ for sufficiently large x . Substituting $u = \log x$,

$$A_n = \int_2^n \frac{\log \log x}{x} dx + O(1) = \log n \log \log n - \log n + O(1) \sim \log n \log \log n.$$

The same last-fraction argument as in Appendix B.5, using $f(\delta n)/f(n) \rightarrow 1$ for fixed $\delta > 0$, gives $W_n \sim n \log \log n$. Hence $r_n \sim (\log n)/n$. Bracketing at $n_{\pm} = (1 \pm \varepsilon)N \log N$ exactly as in Appendix B.5 gives $n_{\text{RV},w} \sim N \log N$ and $n_{\text{RV},w}/n_{\text{pop}} \sim \log N$.

B.7 Geometric-weight alignment and effective averaging

For $w_i = q^i$ with $q > 1$, write $i = n+1-j$. Division by W_n makes the lag j have probabilities proportional to q^{-j} , $j = 0, \dots, n-1$. The denominator $\sum_{j=0}^{n-1} q^{-j} \rightarrow q/(q-1)$ and the numerator for nr_n is

$$\sum_{j=0}^{n-1} q^{-j} \frac{n}{n-j}.$$

For $j \leq n/2$, $n/(n-j) \leq 2$ and tends to 1 for each fixed j ; dominated convergence for the summable geometric series gives the same limit $q/(q-1)$. For $j > n/2$, the sum is at most $n \sum_{j>n/2} q^{-j} \rightarrow 0$.

Thus $nr_n \rightarrow 1$. Direct finite geometric sums also give

$$\frac{\sum_{i=2}^{n+1} q^{2i}}{(\sum_{i=2}^{n+1} q^i)^2} \rightarrow \frac{q-1}{q+1}.$$

Theorem 4.1 supplies the crossing ratio; the displayed square-weight limit measures its bounded effective averaging.

C Near-crossing alignment–concentration obstruction

C.1 The near-crossing scaling

Put $h = \theta^2$. Theorem 3.1 gives $n_{\text{pop}} = h^{-1} + O(1)$. Thus $nh \rightarrow c$ places n at asymptotic fraction c of the population crossing. Moreover $n\Delta_R(n) = 1 - nh \rightarrow 1 - c$, proving the before/after sign interpretation for $c \neq 1$.

C.2 Weight concentration and effective sample size

Let $p_{n,i} = w_i/W_n$, so $\sum_i p_{n,i} = 1$. Alignment is

$$\sum_{i=2}^{n+1} p_{n,i} \left(\frac{n}{i-1} - 1 \right) \rightarrow 0.$$

All summands are nonnegative. For fixed $\varepsilon \in (0, 1)$ and $i - 1 \leq (1 - \varepsilon)n$, the factor is at least $\varepsilon/(1 - \varepsilon)$. Hence the total mass there is at most $(1 - \varepsilon)/\varepsilon$ times the displayed sum, and tends to zero. The remaining set has at most $\varepsilon n + 1$ indices and mass tending to one. Cauchy–Schwarz on that set gives

$$\sum_i p_{n,i}^2 \geq \frac{(\sum_{i-1 > (1-\varepsilon)n} p_{n,i})^2}{\varepsilon n + 1}, \quad \text{so} \quad \liminf_n n \sum_i p_{n,i}^2 \geq \frac{1}{\varepsilon}.$$

For any $K > 0$, choose $\varepsilon < 1/K$; the lower limit is then at least K . Thus $n \sum_i p_{n,i}^2 \rightarrow \infty$.

C.3 Proof of the obstruction theorem

Under alignment, $r_n = (1 + o(1))/n$. If $nh \rightarrow c \neq 1$, then by (2)

$$\frac{\mathbb{E}\Xi_{n,w}}{W_n} = r_n - h = \frac{1 - c + o(1)}{n}.$$

Appendix D.2 proves that the quadratic variance is nonnegative and the additional linear variance equals $4h \sum_i w_i^2$. Therefore

$$\frac{\text{Var}(\Xi_{n,w})}{(\mathbb{E}\Xi_{n,w})^2} \geq \frac{4h \sum_i p_{n,i}^2}{(r_n - h)^2} = \frac{4(c + o(1)) n \sum_i p_{n,i}^2}{(1 - c + o(1))^2} \rightarrow \infty$$

by Appendix C.2. The mean is positive when $c < 1$ and negative when $c > 1$, for sufficiently large n . The proof establishes failure of a sufficient Chebyshev argument, not a positive lower bound on wrong-selection probability. This proves Theorem 5.1.

C.4 The exact boundary

If $nh \rightarrow 1$ under alignment, $r_n - h = o(1/n)$. When this difference is zero, the variance-to-mean-squared ratio is undefined. When it is nonzero, the preceding lower bound has numerator $4h \sum p_{n,i}^2$ and denominator $o(n^{-2})$; writing $r_n - h = \delta_n/n$ with $\delta_n \rightarrow 0$ gives a lower bound $4(1 + o(1))n \sum p_{n,i}^2 / \delta_n^2 \rightarrow \infty$. No leading-order population sign is assigned at $c = 1$.

D Exact variance and a fixed-parameter comparison

D.1 Loss expansion and centering

Write $S_m = \sum_{t=1}^m \xi_t$. Since $\bar{Y}_m = \theta + S_m/m$,

$$(Y_{m+1} - \bar{Y}_m)^2 - Y_{m+1}^2 = (\xi_{m+1} - S_m/m)^2 - (\theta + \xi_{m+1})^2 = \frac{S_m^2}{m^2} - 2\xi_{m+1} \frac{S_m}{m} - 2\theta\xi_{m+1} - \theta^2.$$

The first quadratic term has mean $1/m$ and the other random terms have mean zero. Therefore, with

$$Q_{n,w} = \sum_{m=1}^n w_{m+1} \left\{ \frac{S_m^2}{m^2} - 2\xi_{m+1} \frac{S_m}{m} \right\},$$

we have $\Xi_{n,w} - \mathbb{E}\Xi_{n,w} = Q_{n,w} - 2\theta \sum_{i=2}^{n+1} w_i \xi_i$. This also independently verifies (2).

D.2 Gaussian orthogonality and coefficients

Expand $S_m^2 - m = \sum_{t=1}^m (\xi_t^2 - 1) + 2 \sum_{1 \leq t < u \leq m} \xi_t \xi_u$. Collecting each centered square gives coefficient

$$a_t = \sum_{m=t}^n \frac{w_{m+1}}{m^2}, \quad 1 \leq t \leq n.$$

For each pair $t < l$, the square expansion contributes $2 \sum_{m=l}^n w_{m+1}/m^2$; the validation term at $m = l - 1$ contributes $-2w_l/(l - 1)$. Thus, with empty sums zero,

$$b_l = \sum_{m=l}^n \frac{w_{m+1}}{m^2} - \frac{w_l}{l - 1}, \quad 2 \leq l \leq n + 1,$$

and

$$Q_{n,w} = \sum_{t=1}^n a_t (\xi_t^2 - 1) + 2 \sum_{l=2}^{n+1} b_l \sum_{t=1}^{l-1} \xi_t \xi_l.$$

Independent standard Gaussians satisfy $\mathbb{E}(\xi_t^2 - 1)^2 = 2$ and $\mathbb{E}(\xi_t \xi_l)^2 = 1$ for $t \neq l$. Distinct centered squares, distinct unordered pair products, and a centered square against a pair product have covariance zero: in every cross-product, some independent centered Gaussian appears to an odd power or as a mean-zero factor. Each quadratic monomial also has zero covariance with every ξ_i , because every centered third Gaussian moment is zero. Hence the quadratic and linear parts are orthogonal, and

$$\text{Var}(Q_{n,w}) = 2 \sum_{t=1}^n a_t^2 + 4 \sum_{l=2}^{n+1} (l - 1) b_l^2, \quad \text{Var} \left(-2\theta \sum_{i=2}^{n+1} w_i \xi_i \right) = 4\theta^2 \sum_{i=2}^{n+1} w_i^2.$$

Adding proves the exact identity (5).

D.3 Slow-weight variance orders

Here $h = \theta^2$ and $nh \rightarrow c \in (0, \infty)$; consider $w_{m+1} = f(m)$, where $f(m) = 1, [\log(m+2)]^p$ for fixed $p > 0$, or $\log \log(m+1+e)$. In each case $f(n)$ is slowly varying, and splitting sums at δn as in Appendix B shows

$$\sum_{m=1}^n f(m)^2 \sim n f(n)^2.$$

We verify the quadratic order without borrowing it from 08. The coefficients a_t are bounded by $\sum_{m=t}^{\infty} f(m)/m^2$. For the listed f , integral comparison and eventual monotonicity of $f(x)/x^{3/4}$ give $\sum_{m=t}^{\infty} f(m)/m^2 \leq C f(t)/t$ for large t . Consequently $\sum_t a_t^2 \leq C + C \sum_t f(t)^2/t^2 = O(1)$.

Set $T_n = \sum_{m=n+1}^{\infty} f(m)/m^2$ and $B_l = \sum_{m=l}^{\infty} f(m)/m^2 - f(l-1)/(l-1)$, so $b_l = B_l - T_n$ for $2 \leq l \leq n$ and $b_{n+1} = -f(n)/n$ (the harmless one-step indexing changes the following estimates only by lower-order terms). Integral comparison, summation by parts, and the mean-value theorem give

$$T_n \sim f(n)/n, \quad |B_l| \leq C \left(\frac{\sup_{x \in [l-1, l+1]} |f'(x)|}{1} + \frac{f(l)}{l^2} \right)$$

for smooth extensions of the listed f . To see the derivative bound, subtract the constant- $f(l)$ sum, whose discrepancy from $f(l)/(l-1)$ is $O(f(l)/l^2)$; summation by parts bounds the remaining variation by $C \int_l^{\infty} |f'(x)| x^{-1} dx$. This is $O((\log l)^{p-1}/l)$ for log powers and $O(1/(l \log l))$ for log-log; the constant case has zero derivative. Therefore $D_n = \sum_{l=2}^n (l-1) B_l^2$ is $O(1)$ for $f = 1$ and log-log, while for log powers it is bounded by

$$C + C \sum_{l=3}^n \frac{(\log l)^{2p-2}}{l} = \begin{cases} O(1), & 0 < p < 1/2, \\ O(\log \log n), & p = 1/2, \\ O((\log n)^{2p-1}), & p > 1/2. \end{cases}$$

The three cases follow by $u = \log x$ in the comparison integral $\int^n (\log x)^{2p-2} dx/x$. In every case $D_n = o(f(n)^2)$ when $f(n) \rightarrow \infty$; for $f = 1$, $D_n = O(1)$ and the exact $B_l = O(l^{-2})$ is handled below. The terminal contribution is

$$4T_n^2 \sum_{l=2}^n (l-1) \sim 4 \frac{f(n)^2}{n^2} \frac{n^2}{2} = 2f(n)^2.$$

Expanding $(B_l - T_n)^2$, Cauchy-Schwarz bounds the magnitude of its cross term by $2\{D_n T_n^2 \sum_{l=2}^n (l-1)\}^{1/2} = o(f(n)^2)$ for the growing weights. The end coefficient contributes $4n(f(n)/n)^2 = o(f(n)^2)$. Thus $\text{Var}(Q_{n,w}) \sim 2f(n)^2$ for the growing log and log-log weights. For $f = 1$, direct integral comparison yields $B_l = O(l^{-2})$. Thus $D_n = O(1)$ and $\sum_{l=2}^n (l-1)|B_l| = O(\log n)$. Since $T_n = O(1/n)$, the cross term is $O(\log n/n) = o(1)$; the end coefficient is $O(1/n)$. Therefore $\text{Var}(Q_{n,w}) \sim 2$ also in the unweighted case.

The linear variance is $4h \sum_i w_i^2 \sim 4cf(n)^2$. Hence the full variance is of order $f(n)^2$ (indeed asymptotic to $(2+4c)f(n)^2$ for the growing weights). Meanwhile Appendix B gives $\mathbb{E}\Xi_{n,w} \sim (\log n)f(n)/(p+1)$ for log powers and $\sim (\log n)f(n)$ for log-log; the $-hW_n$ term is only $O(f(n))$. The ordinary-weight mean is $\sim \log n$ and its full variance tends to $2+4c$. Therefore all four Chebyshev ratios have order $(\log n)^{-2}$. Their means are positive for fixed c , establishing the sign statement in the main text.

D.4 Stretched-exponential weights at fixed θ

Let $w_i = \exp(i^\beta)$ with $0 < \beta < 1$. Put $F(x) = e^{x^\beta}$. Substitution $u = x^\beta$ and integration by parts yield, for $a = 1, 2$,

$$\int^n F(x)^a dx = \frac{1}{\beta} \int^{n^\beta} e^{au} u^{1/\beta-1} du \sim \frac{n^{1-\beta} e^{an^\beta}}{a\beta}.$$

Because F is increasing, bounding sums between adjacent integrals changes them by at most $O(F(n)^a)$, negligible relative to $n^{1-\beta} F(n)^a$. Endpoint shifts by one are negligible since $(n+1)^\beta - n^\beta \rightarrow 0$. Therefore

$$W_n \sim \frac{n^{1-\beta} e^{n^\beta}}{\beta}, \quad \sum_i w_i^2 \sim \frac{n^{1-\beta} e^{2n^\beta}}{2\beta}, \quad \frac{\sum_i w_i^2}{W_n^2} \sim \frac{\beta}{2n^{1-\beta}} \rightarrow 0.$$

These sums show an effective terminal window of order $n^{1-\beta} = o(n)$. For any fixed $\delta > 0$, the total weight at $i-1 \leq (1-\delta)n$ is at most $ne^{((1-\delta)n+O(1))^\beta} = o(W_n)$, because the exponential gap is of order n^β . On the remaining indices, $n/(i-1) \leq 1/(1-\delta)$; for the earlier indices their weighted contribution to nr_n is negligible by the same exponential gap even after the factor $n/(i-1) \leq n$. Hence $1 \leq \liminf nr_n \leq \limsup nr_n \leq 1/(1-\delta)$, and $\delta \downarrow 0$ proves alignment.

For fixed $\theta \neq 0$, $\mathbb{E}\Xi_{n,w}/W_n = r_n - \theta^2 \rightarrow -\theta^2$. To bound the quadratic variance, define the centered single-time quadratic increment $U_m = (S_m^2 - m)/m^2 - 2\xi_{m+1}S_m/m$. Gaussian moments and independence give $\|U_m\|_2 \leq \sqrt{2}/m + 2/\sqrt{m} \leq C/\sqrt{m}$. Minkowski's inequality and then Cauchy-Schwarz yield

$$\left\| \frac{Q_{n,w}}{W_n} \right\|_2 \leq C \sum_{m=1}^n \frac{w_{m+1}}{W_n \sqrt{m}} \leq C \left(\sum_{m=1}^n \frac{w_{m+1}}{W_n m} \right)^{1/2} = C \sqrt{r_n} \rightarrow 0.$$

The normalized linear variance is $4\theta^2 \sum_i w_i^2/W_n^2 \rightarrow 0$. Orthogonality from Appendix D.2 then gives $\text{Var}(\Xi_{n,w})/W_n^2 \rightarrow 0$, and division by the mean-squared limit θ^4 proves fixed- θ concentration. This does not apply to $\theta \rightarrow 0$ with $n\theta^2 \rightarrow c$.